SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 3.2: Rotation — Teacher Reference (Pre-filled)

SUMMARY TABLE: MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 3.2: Rotation
Driving Question	DRIVING QUESTION: Why do the blades of the Ngong Hills wind turbine always look the same no matter how far they rotate — and how can we use the mathematics of rotation to predict exactly where any point on the blade will land?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Anchoring Phenomenon & Introduction to Rotation**	Students observed images and/or a short video of the Ngong Hills wind turbines in operation. They noticed that no matter how far the blades spin, the turbine appears visually identical at every position — the blades seem to 'look the same.' The teacher used this as the anchoring phenomenon to surface students' initial questions and prior knowledge about turning and movement.	Students were introduced to the foundational vocabulary of rotation: (1) Centre of rotation — the fixed point about which all turning occurs (analogous to the turbine hub, which stays stationary while blades spin); (2) Angle of rotation — the amount of turn, measured in degrees, from an object point to its image point; (3) Direction of rotation — anticlockwise (positive, the mathematical convention) or clockwise (negative); (4) Object and image — the original figure and its rotated result. Students also learned that rotation is a type of transformation in which every point of a figure turns through the same angle about the same fixed centre.	The Ngong Hills turbine blades 'look the same' after each turn because rotation preserves the shape, size, and orientation of the object — the hub is the centre of rotation, each blade tip traces a circular arc of equal radius, and the three blades are equally spaced at 120° apart, so the overall figure maps onto itself every 120°. Pedagogical note: Use cold-calling (minimum 3 students) to surface misconceptions — a common error is students confusing rotation with reflection or translation. Anchor every definition back to a labelled diagram of the turbine hub and one blade.
Lesson 2 **Properties of Rotation: What Stays the Same and What Changes?**	Students rotated cut-out shapes (or used dynamic geometry software/paper tracing) about a fixed point and compared the object and its image, recording measurements of side lengths, angles, and distances from the centre. They observed that the shape did not change size or form — it simply moved to a new position. They also noticed that the distance from each object point to the	Rotation has four defining properties: (1) The centre of rotation O is fixed — it does not move and is the only invariant point (unless a point lies on O itself); (2) The distance from O to any object point equals the distance from O to its corresponding image point — i.e., $OA = OA'$ for every point A; (3) The angle of rotation is the same for every point — $\angle AOA' = \angle BOB'$ = the	These four properties explain why every Ngong Hills turbine blade tip traces a perfect circle around the hub: each tip is a fixed distance from the hub ($OA = OA'$), and each tip sweeps the same angle per unit time. Pedagogical note: Explicitly contrast with translation (where direction/distance changes position but no fixed centre exists) and reflection (where orientation is

	centre equalled the distance from its image point to the same centre.	stated angle of rotation; (4) The image is congruent to the object — lengths, angles, and area are all preserved. What changes is position (and apparent orientation relative to external reference lines).	reversed). A well-labelled diagram showing OA = OA′ and ∠AOA′ for two different points A and B on the same blade is the key reference diagram for this lesson.
Lesson 3 **Rotating Points on the Cartesian Plane**	Students plotted a single point (e.g., A(3, 2)) on the Cartesian plane, then physically rotated their grid paper (or used tracing paper) by 90°, 180°, and 270° anticlockwise about the origin, recording the new coordinates each time. They observed a predictable numerical pattern in the coordinate pairs as the angle increased.	Students derived and must be able to apply the four coordinate rotation rules about the origin: (1) 90° anticlockwise: (x, y) → (−y, x); (2) 180° (either direction): (x, y) → (−x, −y); (3) 270° anticlockwise (= 90° clockwise): (x, y) → (y, −x); (4) 360°: (x, y) → (x, y) — the point returns to its original position. These rules apply to any point and are derived from the geometry of equal distances from O and equal angles swept. Students should be able to verify each rule using the distance formula: the distance from O must be preserved.	If a blade tip on a Ngong Hills turbine is located at coordinates (3, 2) relative to the hub at the origin, after a 90° anticlockwise rotation it will be at (−2, 3), after 180° at (−3, −2), and after 270° anticlockwise at (2, −3). The rules allow engineers and mathematicians to predict exact positions without measuring every time. Pedagogical note: Require students to verify at least one transformation using the distance formula $d = \sqrt{(x^2 + y^2)}$ to confirm the distance from O is unchanged. Cold-call at least 4 students to state a rule unprompted.
Lesson 4 **Rotating Shapes on the Cartesian Plane**	Students plotted a triangle or quadrilateral with labelled vertices on the Cartesian plane and applied the coordinate rotation rules (from Lesson 3) to each vertex individually, then connected the image vertices to form the rotated shape. They observed that the image shape was congruent to the object and that each vertex moved by the same angle about the origin.	To rotate a shape about the origin, students apply the appropriate coordinate rule to every vertex of the shape: 90° anticlockwise maps (x, y) to (−y, x); 180° maps (x, y) to (−x, −y); 270° anticlockwise maps (x, y) to (y, −x). The image vertices are then joined in order to produce the rotated shape. Key procedural steps: (1) List all vertices; (2) Apply the rule to each vertex; (3) Plot image vertices; (4) Connect image vertices in the same order as the object; (5) Label the image with prime notation (A′, B′, C′). The image is always congruent to the object.	Rotating a full turbine blade shape (modelled as a polygon) about the hub (origin) by 120° anticlockwise produces an image that perfectly overlaps the position of the next blade — demonstrating both the mathematics of the rotation rules and the visual 'sameness' of the turbine. Pedagogical note: A common student error is applying the rule to only some vertices or mis-signing the coordinates. Insist students show a two-column table of object coordinates → image coordinates before plotting. Use a worked example with a triangle representing a simplified turbine blade cross-section.
Lesson 5 **Angle and Centre of Rotation — Finding What We Cannot See**	Students were given an object shape and its image on the Cartesian plane (or dot paper) without being told the centre or angle of rotation. They attempted to locate the centre by inspection, then used a ruler and compass (or set square) to find the perpendicular bisectors of lines joining corresponding object–image point pairs.	Students learned two key constructions: (1) Finding the centre of rotation — construct the perpendicular bisector of the segment joining each object point to its corresponding image point; the perpendicular bisectors of at least two such segments intersect at the centre of rotation O. (2) Finding the angle of rotation — once O is located, join any object point A to O and its image A′ to O; the angle ∠AOA′ (measured with a protractor) is the angle of rotation. Direction (clockwise or	If a wind-monitoring engineer at Ngong Hills photographs the turbine at two moments and needs to know exactly how far it has rotated (and about which precise hub point), they would use exactly these constructions: perpendicular bisectors to locate the hub and angle measurement to determine rotation. This shows that mathematics can recover 'hidden' information from observable outcomes. Pedagogical note: Allow adequate wait time (minimum 30 seconds) for students to attempt the

		anticlockwise) is determined by noting which way A moved to reach A′ around O.	construction before demonstrating. Emphasise that two perpendicular bisectors are the minimum needed; a third provides a verification check.
Lesson 6 **Order of Rotational Symmetry**	Students were given cut-out regular polygons (equilateral triangle, square, regular pentagon, regular hexagon) and common Kenyan objects/symbols (e.g., the Kenyan shilling coin, a maize cross-section, a traditional Maasai shield design) and rotated them about their centres, counting how many times the shape mapped exactly onto itself before completing a full 360° turn.	A shape has rotational symmetry of order n if it maps exactly onto itself n times during a full 360° rotation. The minimum angle of rotation that produces such a mapping is the angle of rotational symmetry = 360° ÷ n. Key results: equilateral triangle — order 3, minimum angle 120°; square — order 4, minimum angle 90°; regular pentagon — order 5, minimum angle 72°; regular hexagon — order 6, minimum angle 60°. Every shape has at least order 1 (a full 360° rotation). A shape with order 1 only is said to have no rotational symmetry. The three-bladed Ngong Hills turbine has rotational symmetry of order 3 (minimum angle 120°).	The Ngong Hills wind turbine 'always looks the same' precisely because it has rotational symmetry of order 3: every 120° rotation maps the turbine onto itself. This is the mathematical answer to the anchoring phenomenon. Pedagogical note: Connect explicitly to the driving question here — this is a pivotal lesson. Have students record on the Driving Question Board: 'The turbine has order-3 rotational symmetry; minimum rotation angle = 120°.' Use the maize cross-section as a local food-context example (typically order 8–16 depending on kernel rows) to extend thinking.
Lesson 7 **Rotation About a Point Other Than the Origin**	Students plotted a shape and a centre of rotation C that was NOT at the origin (e.g., C(2, 1)) and attempted to rotate the shape. They observed that simply applying the origin-based coordinate rules gave the wrong image — the shape did not rotate correctly about C. This productive struggle helped students see the need for a new approach.	To rotate a shape about a point C(a, b) other than the origin: (1) Translate the entire figure so that C moves to the origin — subtract (a, b) from every vertex: (x, y) → (x − a, y − b); (2) Apply the standard origin rotation rule for the required angle; (3) Translate back by adding (a, b) to every image vertex: (x′, y′) → (x′ + a, y′ + b). This three-step process (translate → rotate → translate back) is the general method. Students should present working in a three-column table: original coordinates | shifted coordinates | final image coordinates.	The turbine hub at Ngong Hills is not always placed at a coordinate origin in engineering drawings — it may be at any reference point on a grid. Using the translate–rotate–translate method, engineers can calculate exact blade tip positions regardless of where the hub appears on the plan. Pedagogical note: A common error is forgetting to translate back at step 3. Insist on the three-column table as a non-negotiable written format. Use C(2, 1) with a triangle representing a blade cross-section as the class worked example.
Lesson 8 **Congruence and Rotation: Proving the Turbine Blades Are Congruent**	Students measured and compared corresponding sides and angles of an object shape and its rotated image (from earlier lessons), recording results in a table. They observed that every pair of corresponding lengths was equal and every pair of corresponding angles was equal, regardless of the angle of rotation applied.	Rotation is an isometry — a transformation that preserves length, angle size, and area — making the image congruent to the original object. Formally: if shape A′B′C′ is the image of shape ABC under any rotation, then A′B′ = AB, B′C′ = BC, A′C′ = AC, and all corresponding angles are equal, therefore △ABC ≅ △A′B′C′ (by SSS or SAS congruence). The three properties preserved are: (1) side length; (2) angle measure; (3) area. Orientation is preserved (unlike reflection, which reverses orientation). Students must be able to state	All three Ngong Hills turbine blades are congruent to each other because each blade is simply a rotation of the first blade by 120° and 240° about the hub. Because rotation is an isometry, every dimension of every blade is identical — this is essential for mechanical balance and consistent energy generation. Pedagogical note: Use the formal congruence notation △ABC ≅ △A′B′C′ and require students to justify each corresponding part. Contrast with dilation (which is NOT an isometry) to sharpen understanding of what 'isometry' means.

		a formal congruence statement with correct vertex correspondence.	
Lesson 9 **Applications of Rotation in Real-Life Situations**	Students examined three real-life rotation contexts: (1) an analogue clock face, identifying the angle swept by the minute and hour hands over various time intervals; (2) a Ferris wheel at the Nairobi Uhuru Gardens, analysing passenger gondola positions after partial turns; (3) a rotating Kenyan chapati (flat bread) on a pan, discussing symmetry and even cooking. Students measured or calculated angles and matched them to coordinate transformations.	Key application results: (1) Clock hands — the minute hand sweeps 360° ÷ 60 min = 6° per minute (clockwise, therefore negative by convention); the hour hand sweeps 360° ÷ 12 hr = 0.5° per minute. (2) Ferris wheel — a gondola's position after a rotation of θ° about the wheel's fixed centre can be computed using the rotation coordinate rules; at 90° anticlockwise from the rightmost point, the gondola is at the topmost point. (3) General principle — any real-world object turning about a fixed axis obeys the same rotation mathematics. Students should be able to set up and solve angle-of-rotation problems in context.	The same mathematics that describes a Ngong Hills turbine blade also describes a Nairobi Ferris wheel gondola, a clock hand, and a chapati rotating on a pan — all are rotations about a fixed centre. Recognising rotation in real life allows engineers, designers, and everyday users to make accurate predictions about position and timing. Pedagogical note: Cold-call at least 3 students to identify the centre, angle, and direction in each new context before working the calculation. Emphasise unit consistency — angles must be in degrees unless otherwise specified at this level.
Lesson 10 **Unit Consolidation, Model Finalisation & DQB Review — Answering the Driving Question**	Students reviewed all entries on the Driving Question Board (DQB) accumulated across Lessons 1–9, identifying which questions had been answered, which had been partially answered, and whether any new questions remained. They also revisited the anchoring phenomenon — the Ngong Hills wind turbine — and compared their Lesson 1 initial model/explanation with their current, fully developed model.	The complete mathematical answer to the driving question integrates all unit learning: (1) The turbine blades 'always look the same' because the three-bladed rotor has rotational symmetry of order 3 — it maps onto itself every 120°; (2) The hub is the centre of rotation — a fixed invariant point; (3) Rotation preserves length, angle, and area (isometry), so all three blades are congruent; (4) Coordinate rotation rules — $(x,y)\rightarrow(-y,x)$ for 90° anticlockwise, $(x,y)\rightarrow(-x,-y)$ for 180°, $(x,y)\rightarrow(y,-x)$ for 270° anticlockwise — allow exact prediction of any blade-tip position; (5) For a hub not at the origin, the translate–rotate–translate method extends these rules. Together, these ideas form a complete, coherent model of rotation.	The driving question is fully answered: the Ngong Hills turbine blades always look the same because the rotor has order-3 rotational symmetry (minimum angle 120°), and we can predict any blade-tip position exactly using the coordinate rotation rules together with the properties of isometry. This lesson closes the learning arc that began with student wonder in Lesson 1 and built concept by concept to a rigorous mathematical model. Pedagogical note: Allocate time for students to write or present their final model explanation in their own words — this is the assessment-for-learning capstone. Celebrate the journey from 'I wonder why…' to 'I can prove that…' as a reflection on mathematical thinking.